

India's Climate and Environmental Transition

Emissions, air quality, the energy transition and the agricultural pressures behind them.

4 CHAPTERS	24 INDICATORS	5 WITH STATE DATA	1951–2025 PERIOD COVERED
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Generated 21 Aug 2026 • assembled from 22 sources

EXECUTIVE SUMMARY

This report assembles 24 indicators across 4 chapters, drawn from 22 sources and covering 1951–2025.

Of those indicators, 5 carry state-level detail and can be mapped across India's states. The remaining 19 are published at national level only by their source, and are shown here as national series without a sub-national breakdown.

Coverage limits. This report contains no district- or city-level figures. Comparable district data is not currently integrated into India Insight Lab, and no value here has been estimated, interpolated or carried down from a higher geography to fill that gap. Where a number is absent, it is absent.

KEY FINDINGS

Each finding below is computed directly from the state values in this report — the highest and lowest recorded state for that indicator. None are authored interpretations.

1 Forest cover by state

Ranges from 85.4% in Mizoram to 3.6% in Haryana — a spread of 81.8% across the 8 states reported.

India State of Forest Report, Forest Survey of India

2

Slum population share by state

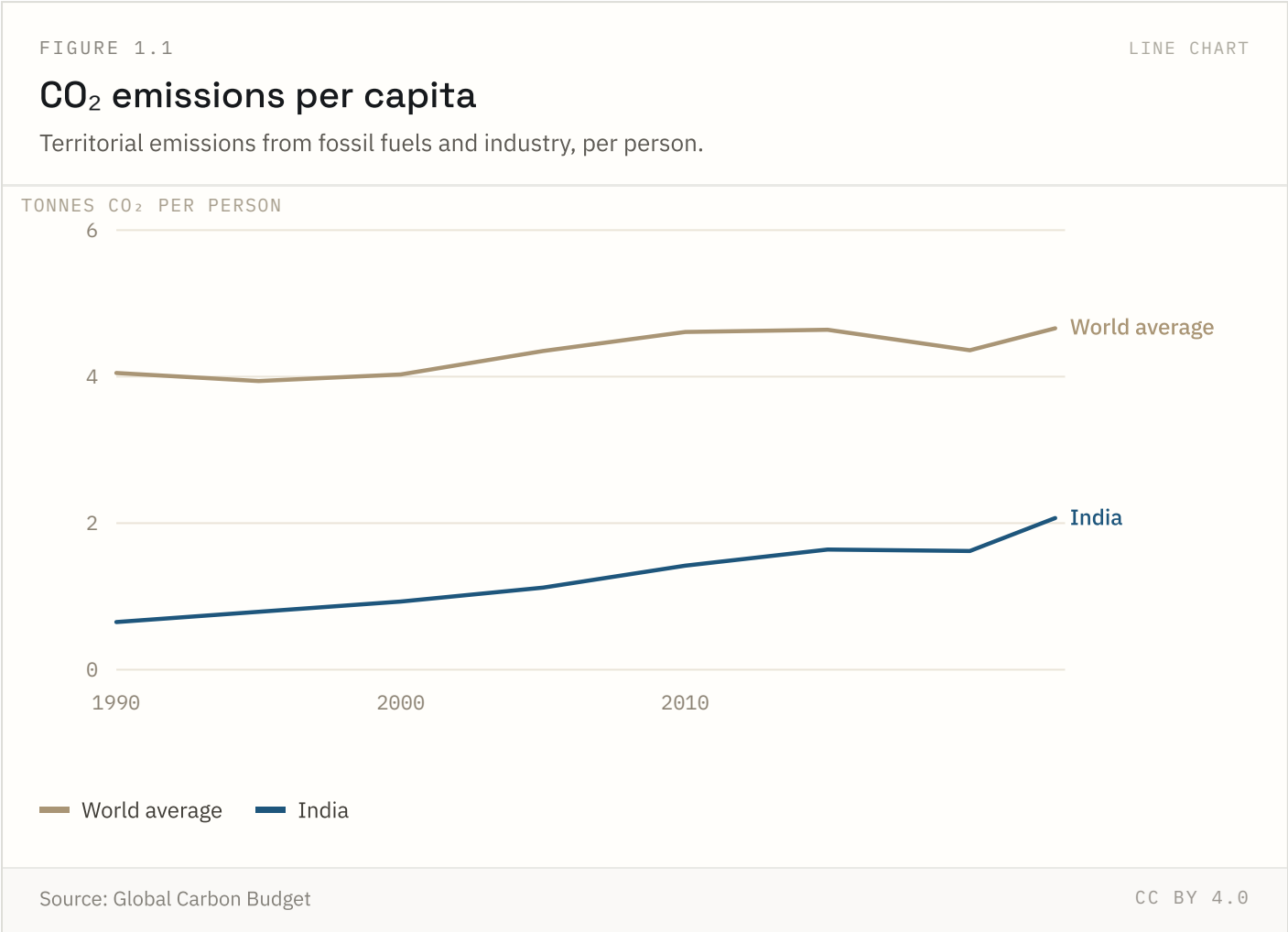
Ranges from 24.0% in Andhra Pradesh to 4.9% in Kerala — a spread of 19.1% across the 15 states reported.

Census of India 2011, Primary Census Abstract for Slum

Climate & air quality

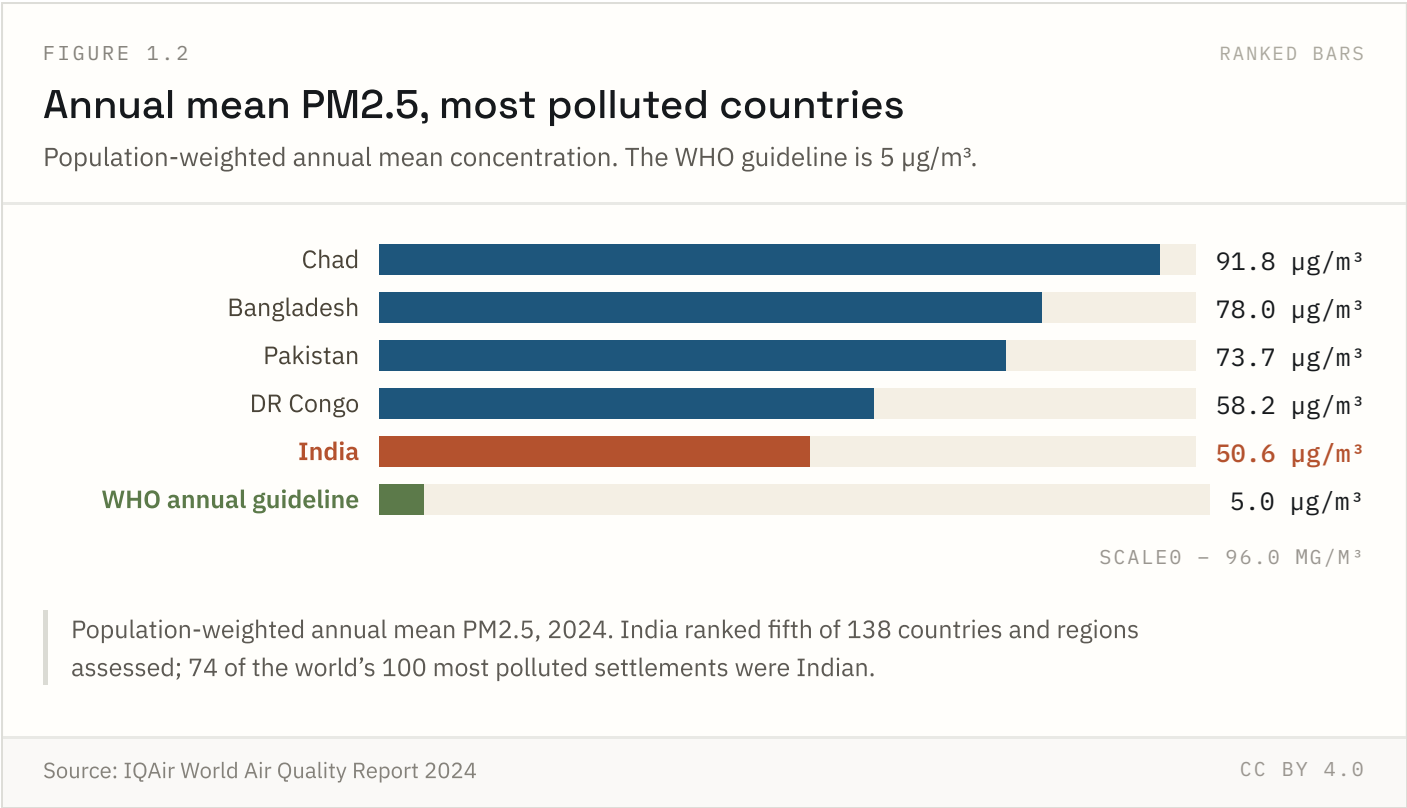
1 of 10 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 1.1 NATIONAL ONLY



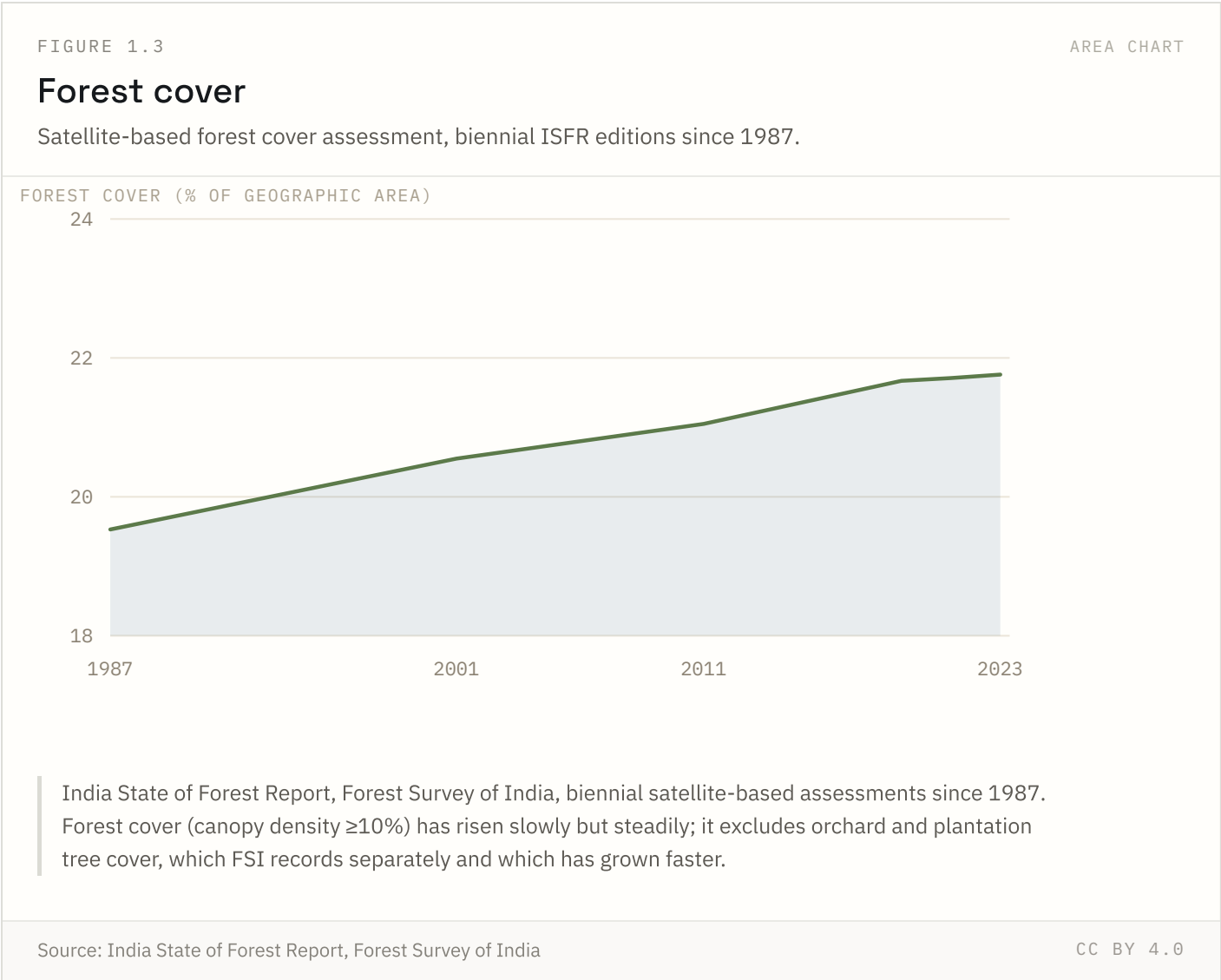
India is the third-largest emitter in absolute terms and among the lowest per person of any major economy — roughly 45 per cent of the world average. Cumulative emissions since 1850 are about 3 per cent of the global total.

FIGURE 1.2 NATIONAL ONLY



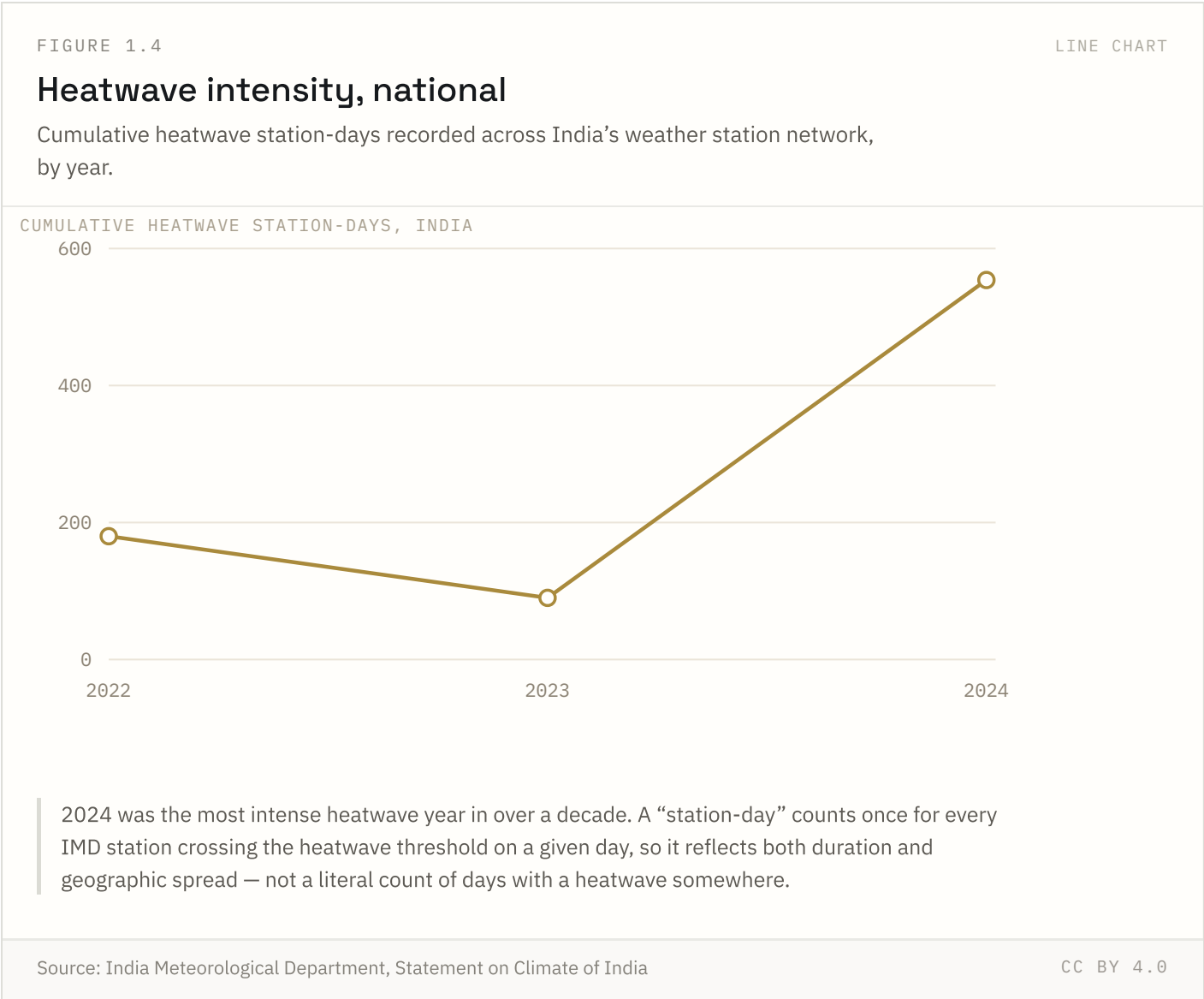
Every city shown exceeds the WHO annual guideline of 5 µg/m³, and all but one exceed India’s own standard of 40. Monitor coverage is uneven: much of central and eastern India has fewer than one continuous monitor per million people.

Source: IQAir World Air Quality Report 2024 Coverage: 2024 Updated: 14 Apr 2026



Forest cover has risen slowly but steadily, from 19.53% of geographic area in 1987 to 21.76% in 2023 — still short of the National Forest Policy target of 33%. The figure excludes orchard and plantation tree cover, tracked separately, which has grown faster than natural forest.

FIGURE 1.4 NATIONAL ONLY

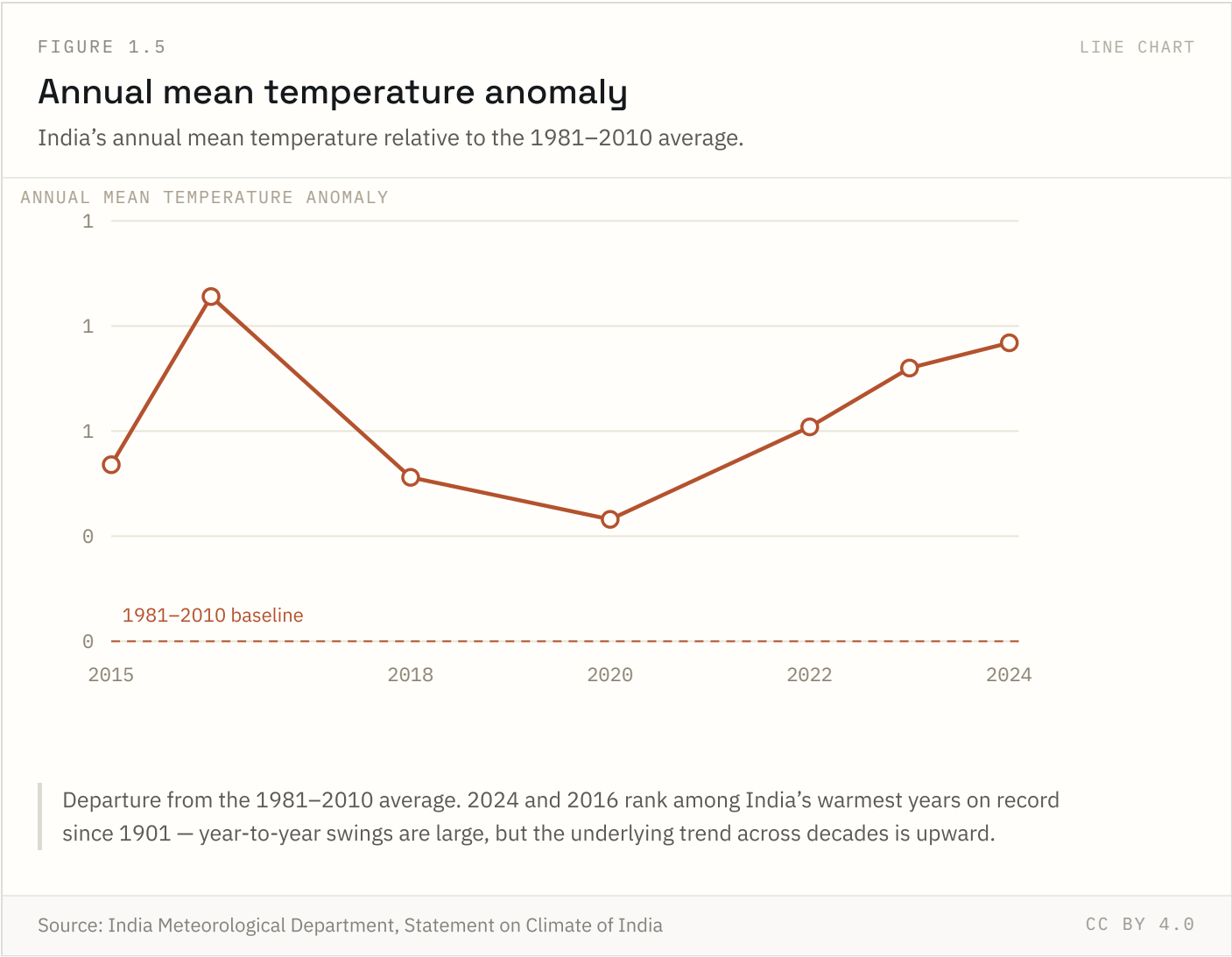


2024 was widely reported as India’s most intense heatwave year in over a decade, with several cities recording temperatures above 45°C. The station-day metric rewards both longer heatwaves and wider geographic spread, so a single very hot, very long event in a densely monitored region can dominate the national total.

Source: India Meteorological Department, Statement on Climate of India Coverage: 2022–2024

Updated: 18 Aug 2026

FIGURE 1.5 NATIONAL ONLY

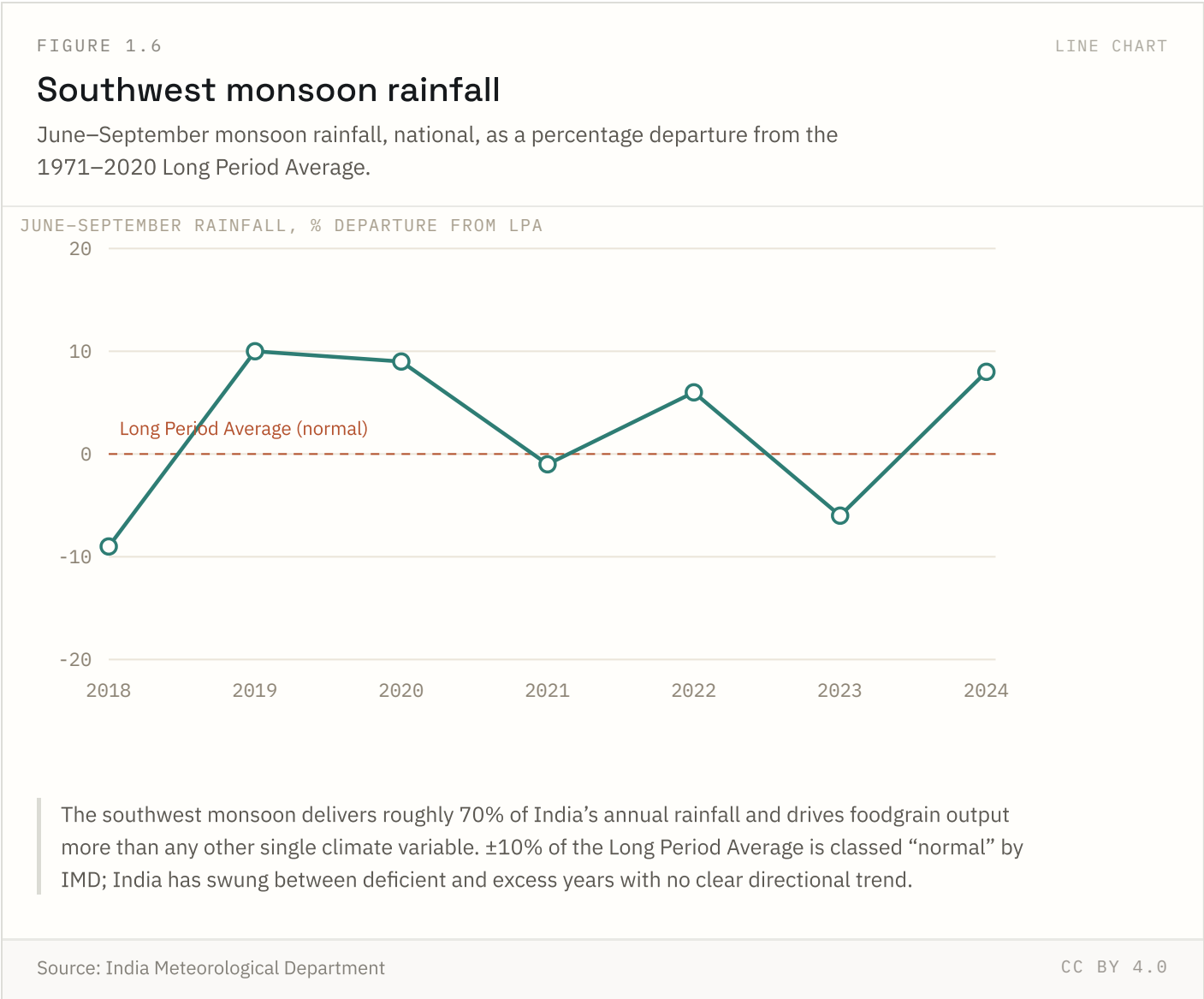


IMD has ranked several recent years among the warmest since national records began in 1901. Annual figures are noisy year to year — a single cool year does not reverse the multi-decade warming trend visible in the underlying data.

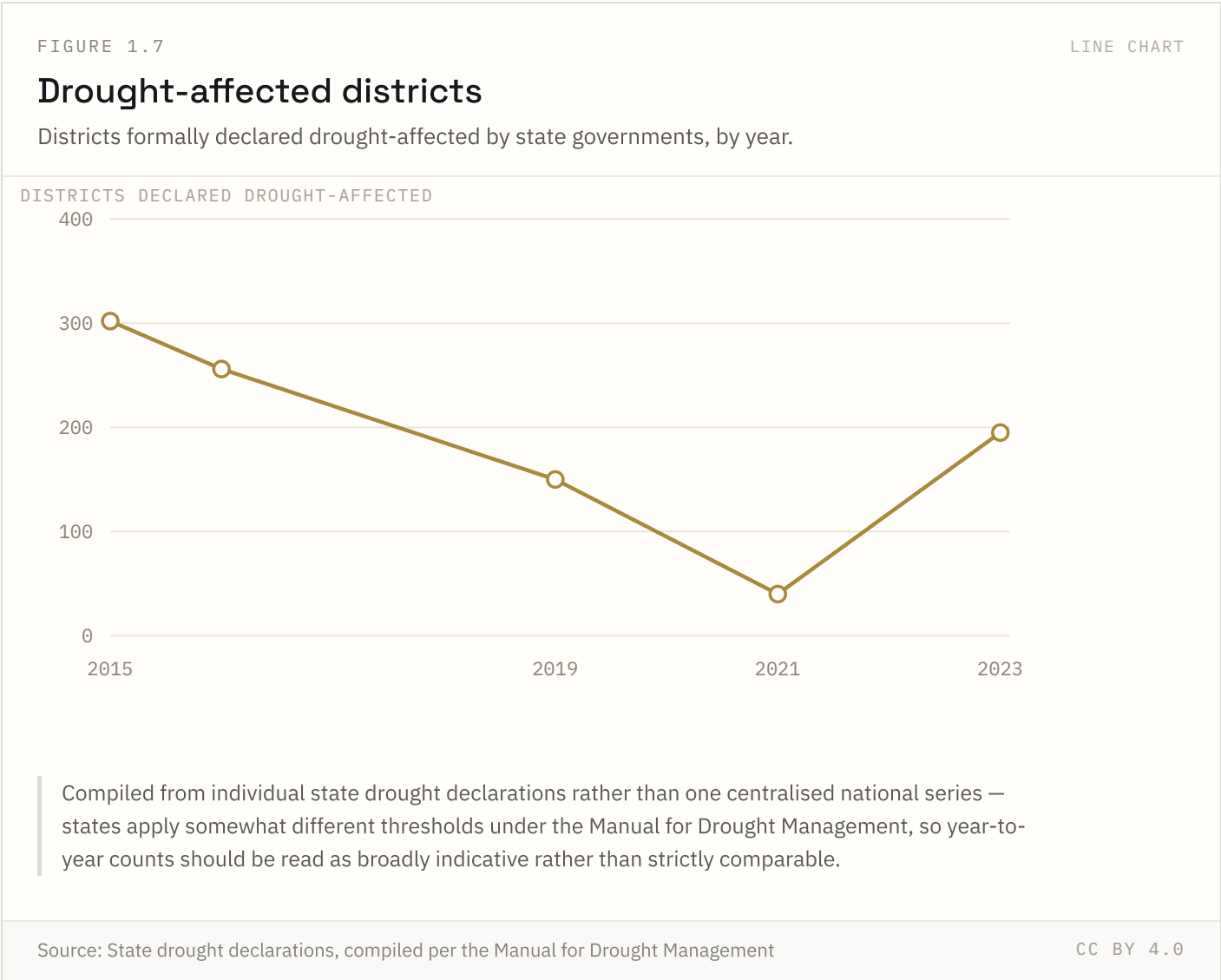
Source: India Meteorological Department, Statement on Climate of India Coverage: 2015–2024

Updated: 18 Aug 2026

FIGURE 1.6 NATIONAL ONLY



IMD classes $\pm 10\%$ of the LPA as “normal.” The monsoon’s national total can mask sharply uneven regional distribution — a “normal” year on paper can still combine drought in one region with flooding in another.

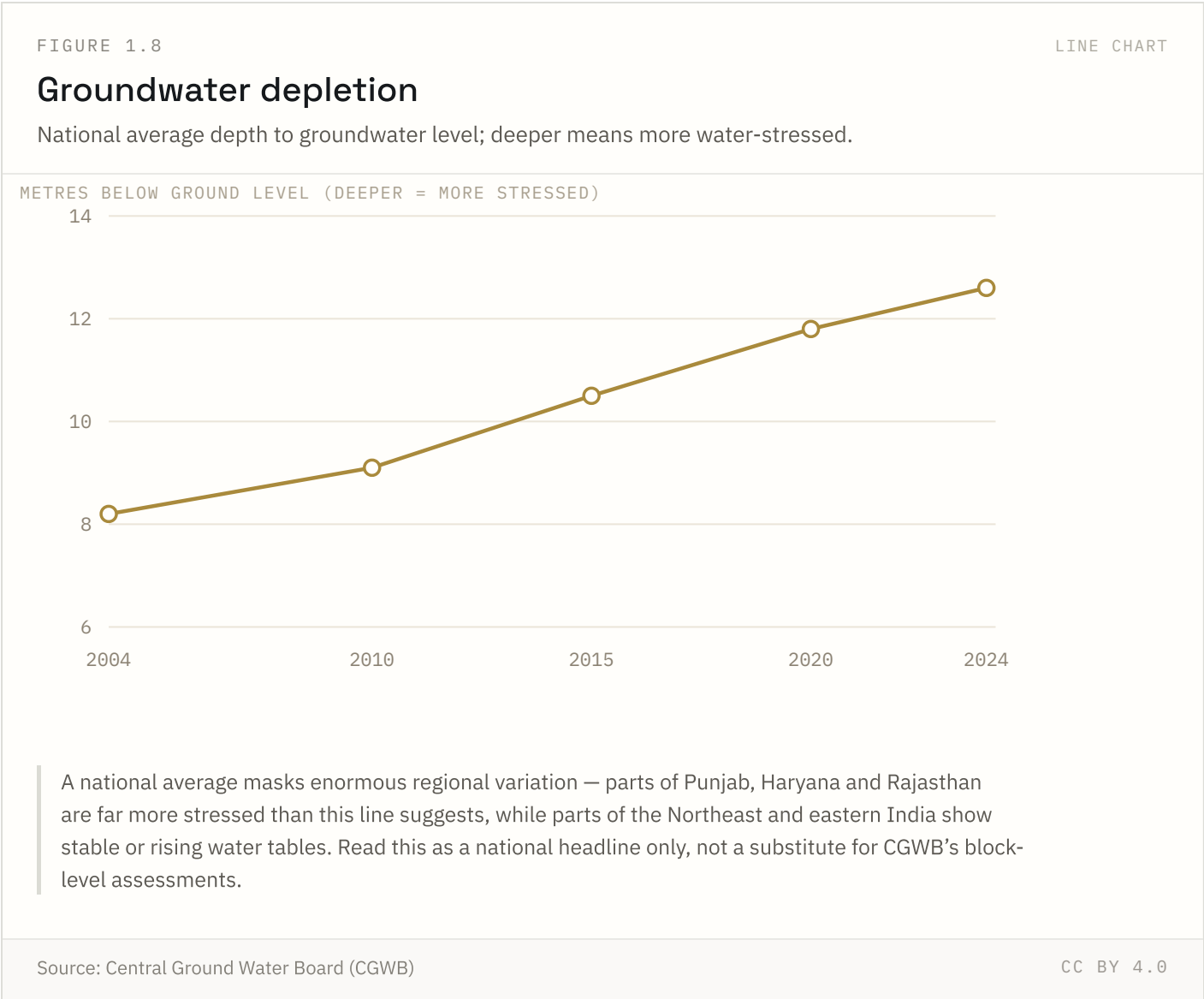


There is no single centralised national drought register — this compiles individual state declarations, which apply the Manual for Drought Management’s criteria with some local discretion. Treat trends as broadly indicative; exact counts vary by source and reporting date.

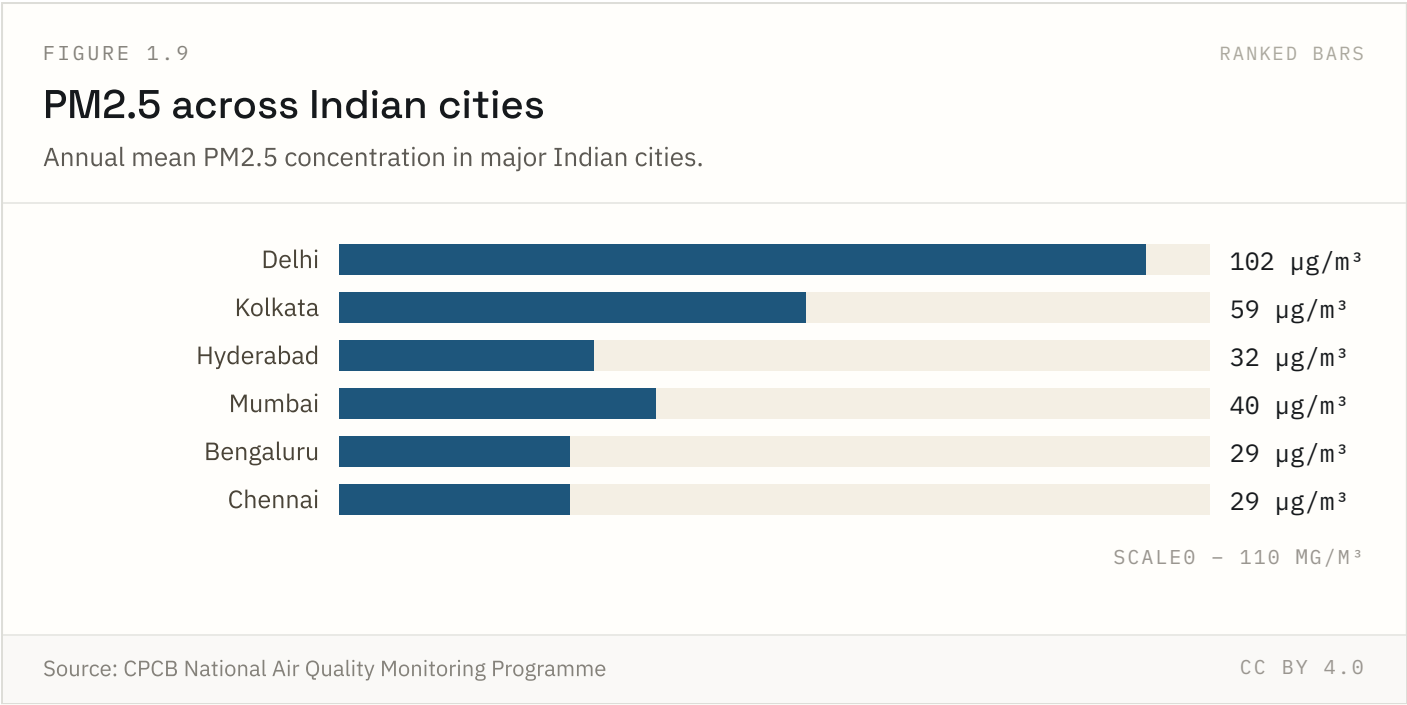
Source: State drought declarations, compiled per the Manual for Drought Management Coverage: 2015–2023

Updated: 18 Aug 2026

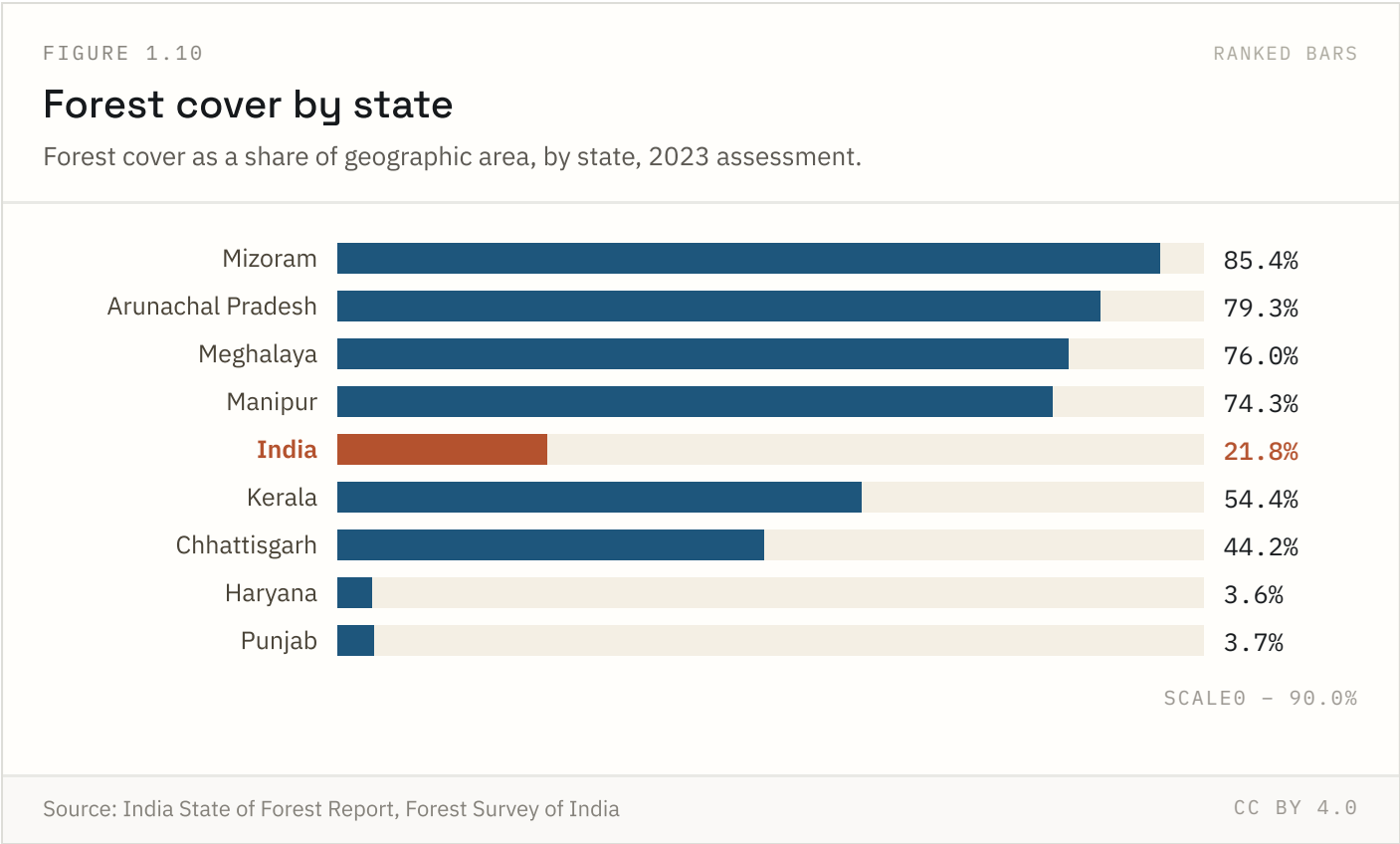
FIGURE 1.8 NATIONAL ONLY



India is the world’s largest extractor of groundwater, used mainly for irrigation. A national average conceals sharp regional divergence — Punjab, Haryana and Rajasthan are critically over-extracted, while parts of eastern India show stable or even rising water tables.



Every city shown exceeds both the WHO guideline (5 $\mu\text{g}/\text{m}^3$) and India’s own national standard (40 $\mu\text{g}/\text{m}^3$). Delhi’s winter inversion and regional stubble-burning make its annual average an outlier even among India’s other large, heavily polluted cities.



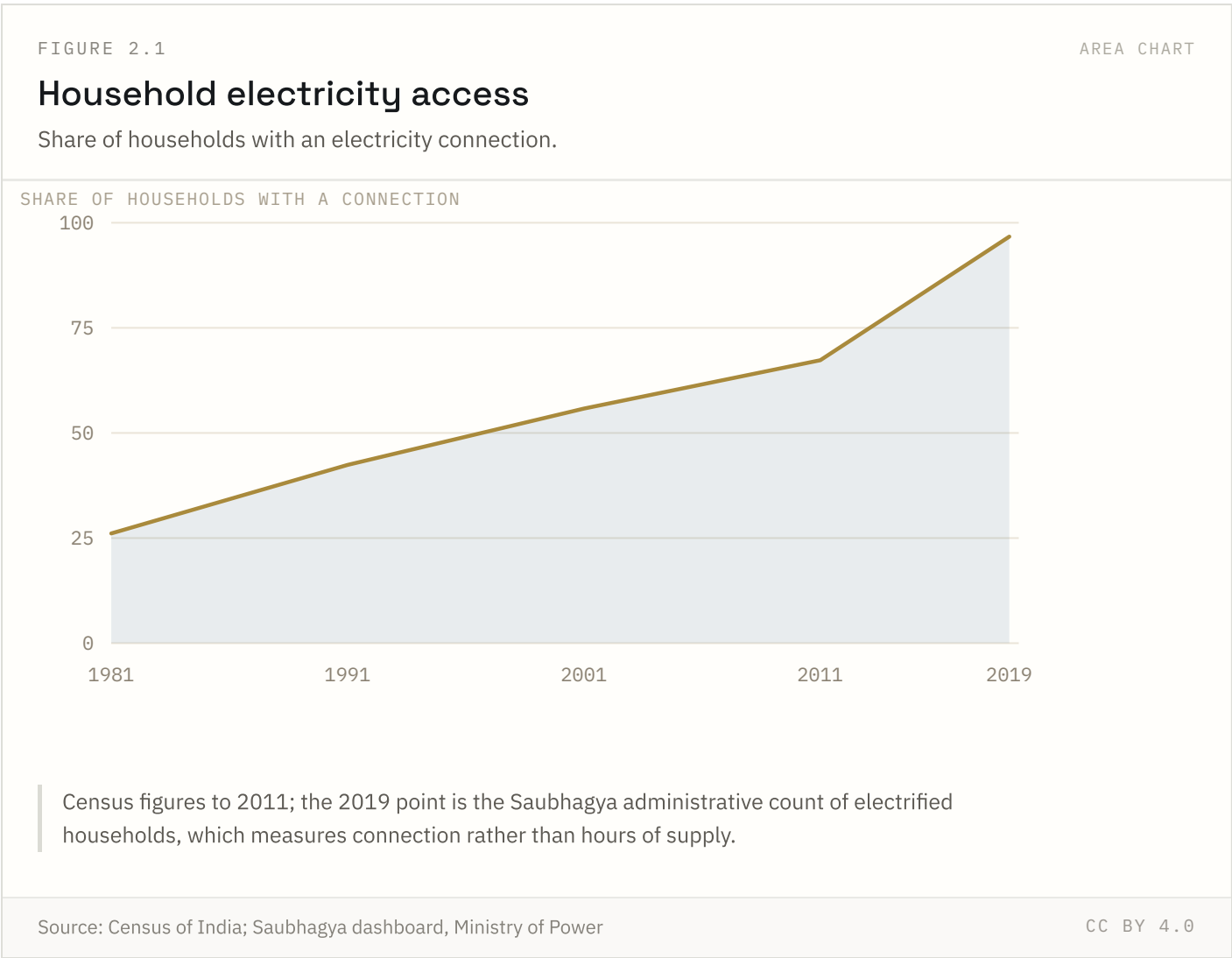
The gap between India’s most and least forested major states spans more than 80 percentage points. Northeastern states carry a hugely disproportionate share of India’s forest cover relative to their land area’s share of the country.

Highest: **Mizoram** at 85.4%. Lowest: **Haryana** at 3.6%.

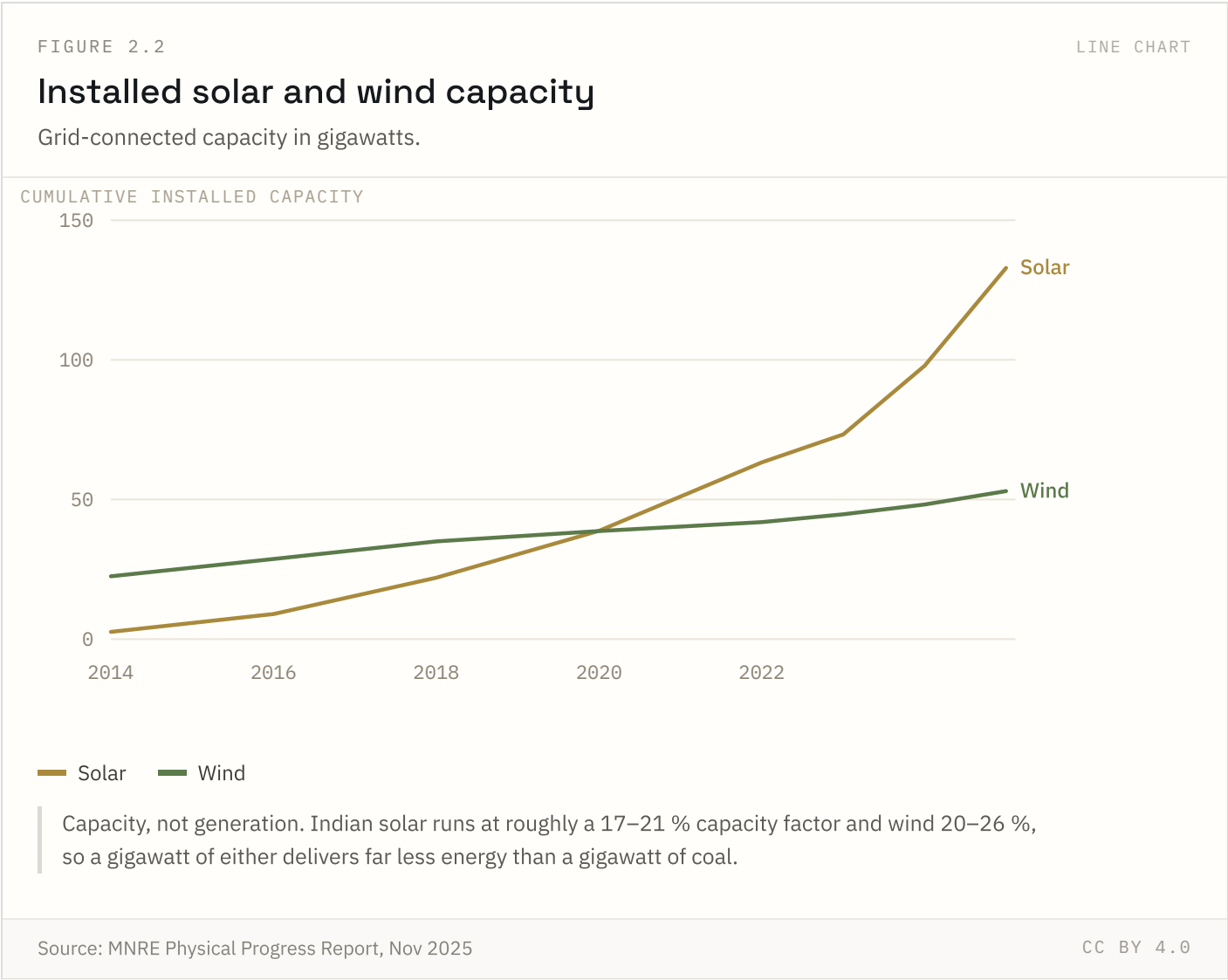
Energy & electricity

2 of 5 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 2.1 NATIONAL + STATE

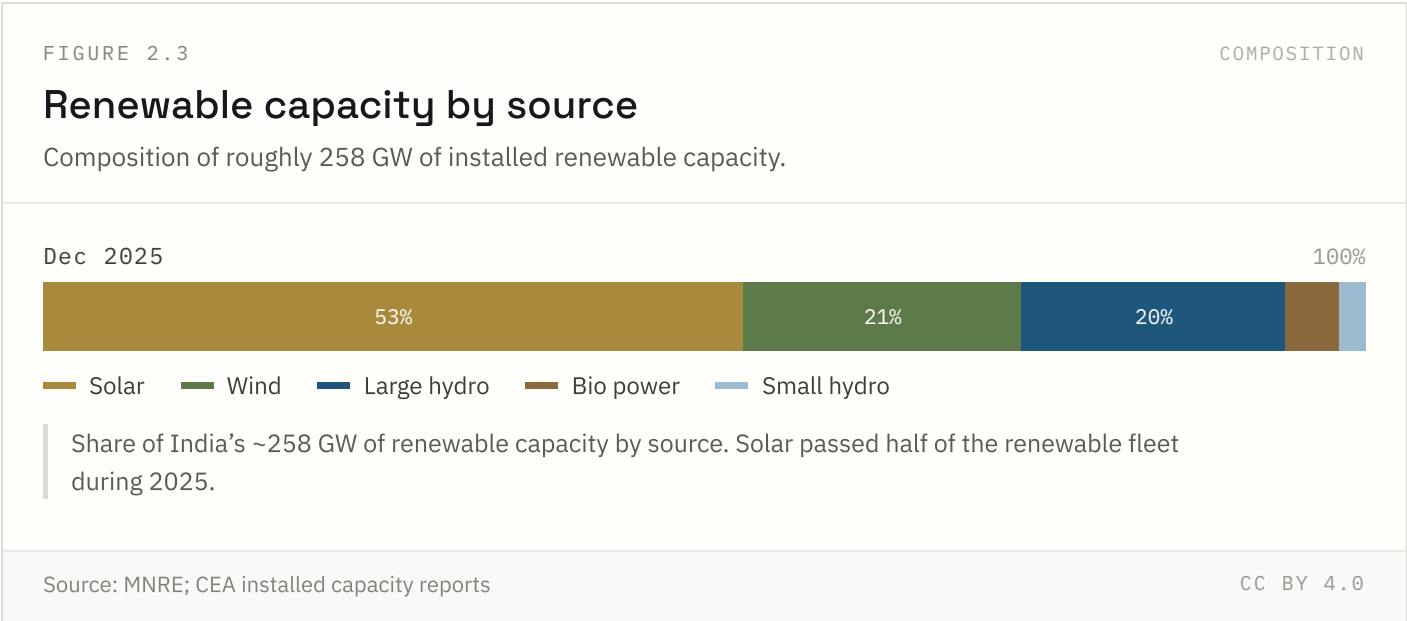


Connection is not the same as supply. Administrative data reports near-universal connection since 2019, but survey data on hours of supply shows large rural shortfalls, especially in Bihar, Jharkhand and Uttar Pradesh.



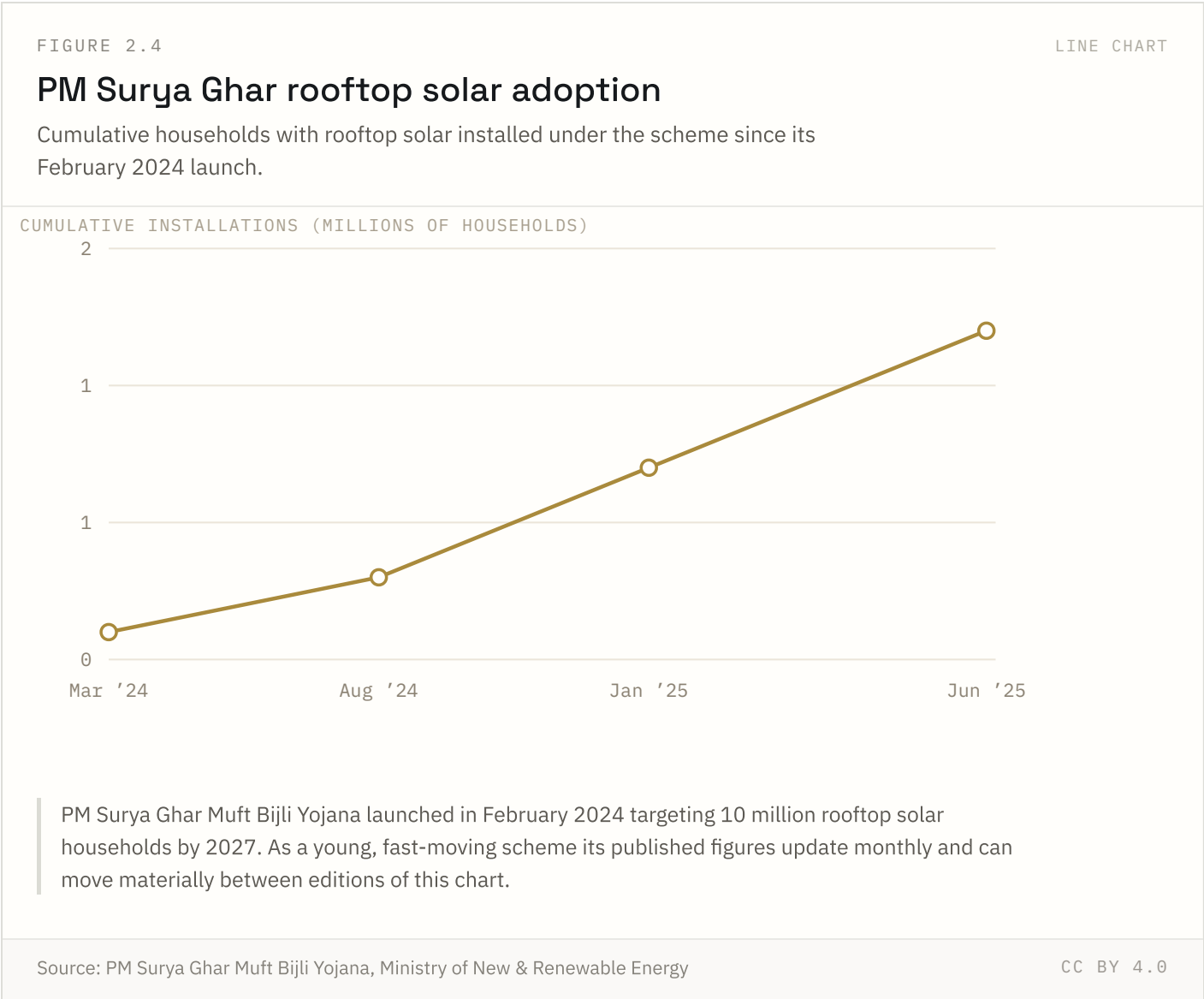
Capacity is not generation. Solar capacity factors in India run 17–21 per cent, wind 20–26 per cent, so a gigawatt of each delivers far less energy than a gigawatt of coal. Both series are ahead of the trajectory implied by the 2015 targets.

FIGURE 2.3 NATIONAL ONLY



Solar passed half the renewable fleet during 2025. Large hydro, which took seven decades to build, is now smaller than solar, which took eleven years.

Source: MNRE; CEA installed capacity reports Coverage: Dec 2025 Updated: 30 May 2026

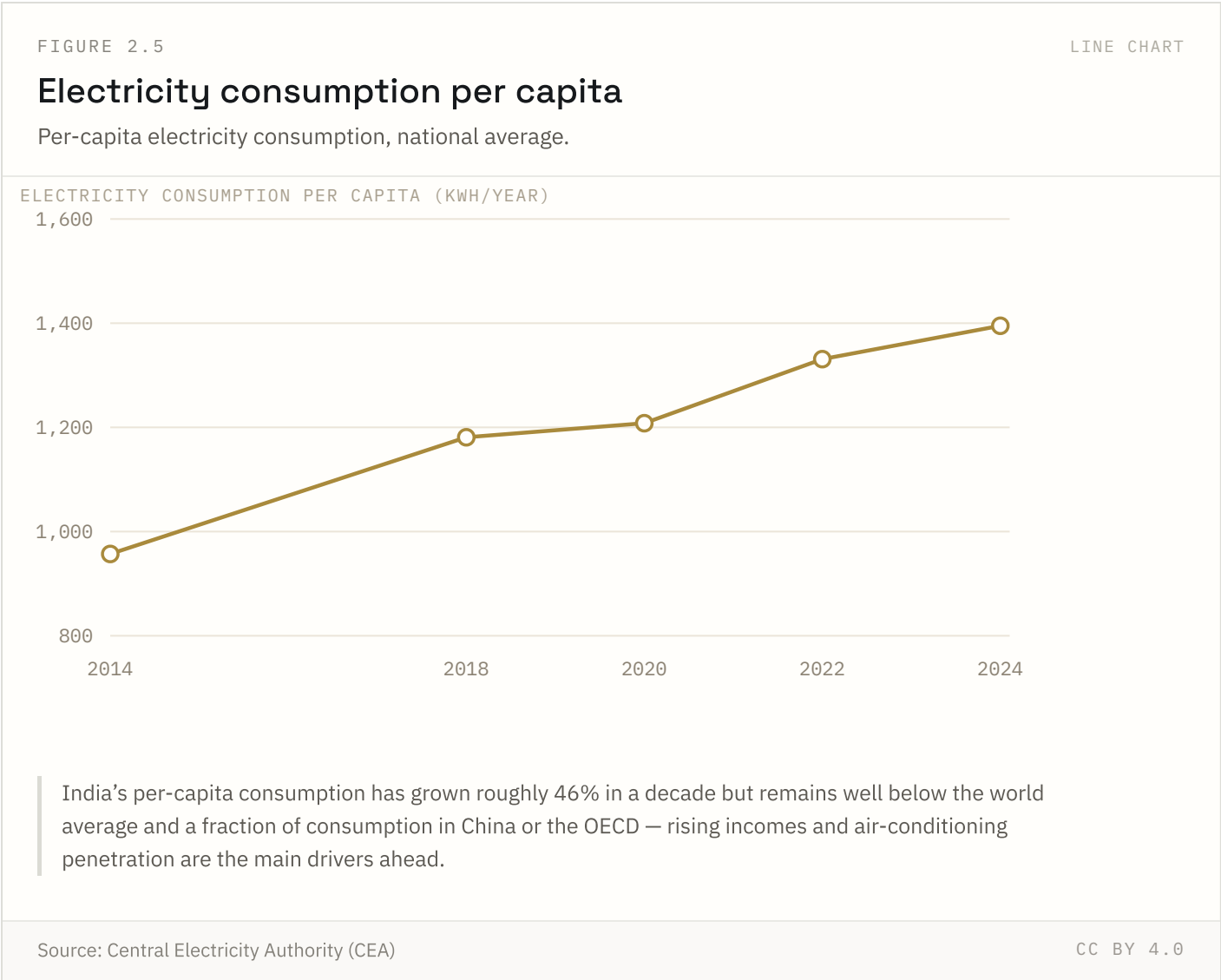


The scheme targets 10 million rooftop-solar households by 2027; adoption so far is a small fraction of that target and figures update frequently as installation data is reconciled with subsidy disbursement.

Source: PM Surya Ghar Muft Bijli Yojana, Ministry of New & Renewable Energy Coverage: Mar 2024–Jun 2025

Updated: 18 Aug 2026

FIGURE 2.5 NATIONAL ONLY



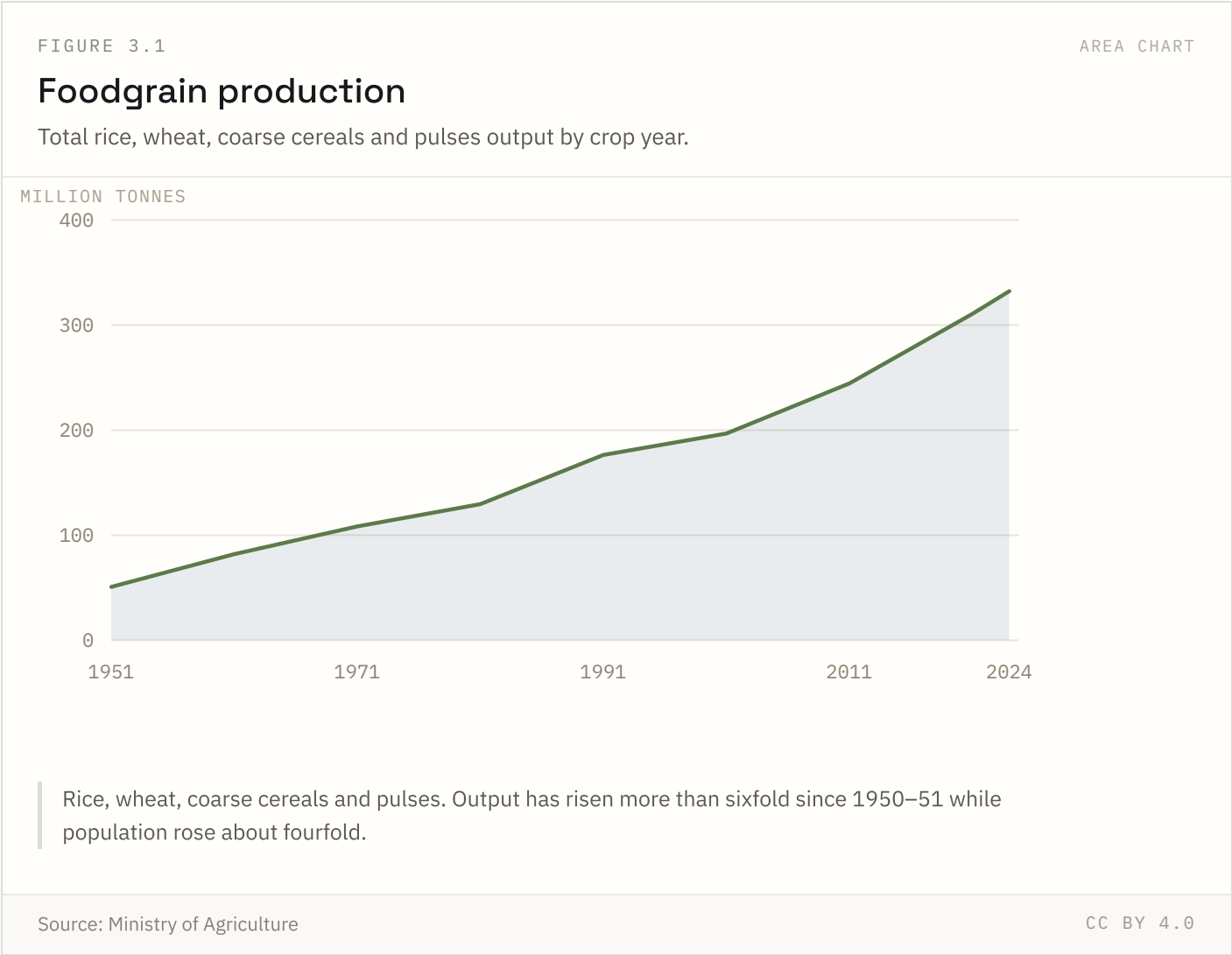
This is a different question from your household electricity access indicator: access asks whether a connection exists; this asks how much power the average person actually uses — a better proxy for living standards and industrial activity than connection rates alone.

Source: Central Electricity Authority (CEA) Coverage: 2014–2024 Updated: 18 Aug 2026

Agriculture & food

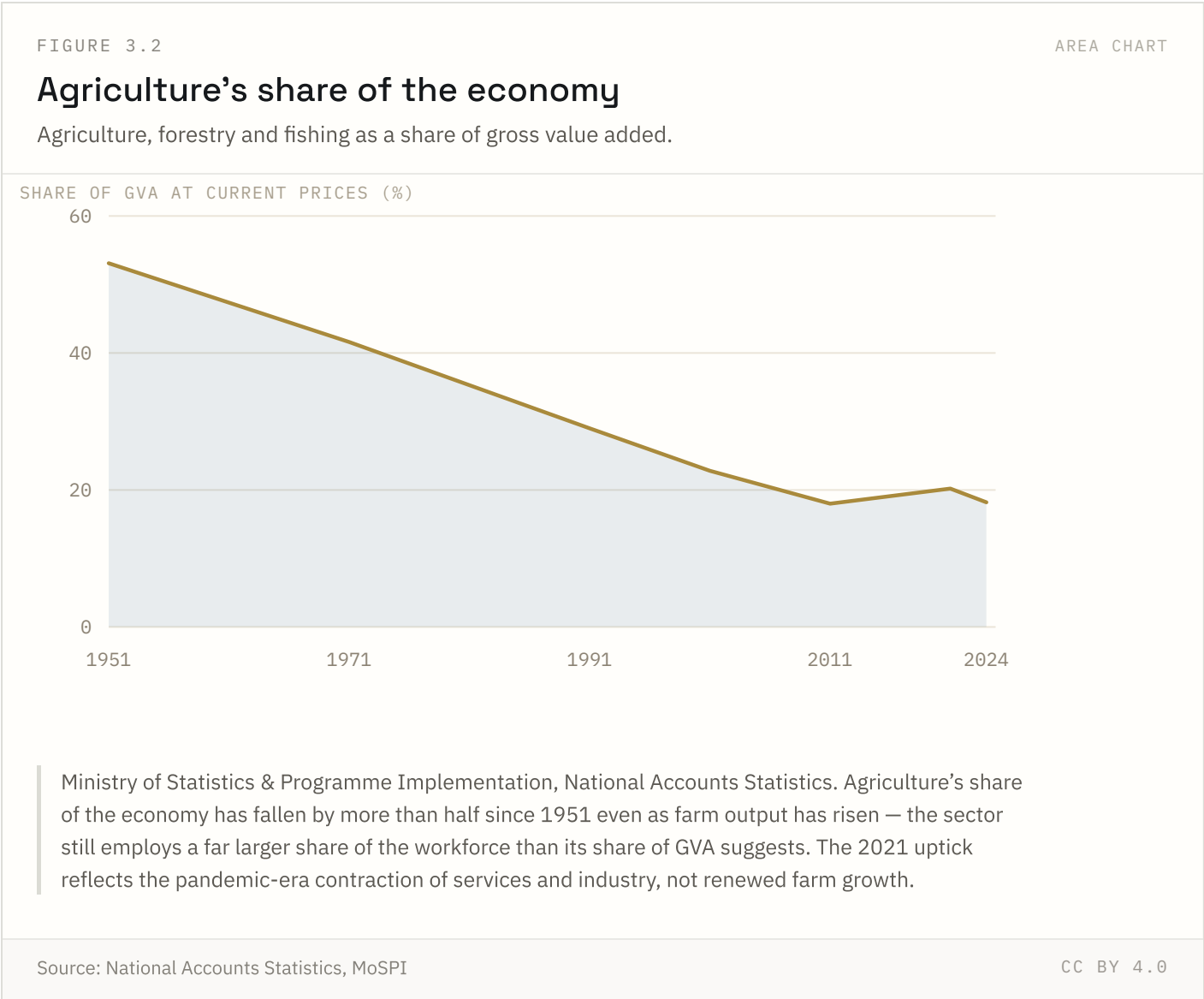
1 of 2 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 3.1 NATIONAL + STATE



Output has risen more than sixfold since 1951 while the population rose about fourfold. Yield growth, not area expansion, accounts for nearly all of the gain since 1980 – and it has slowed markedly since 2015.

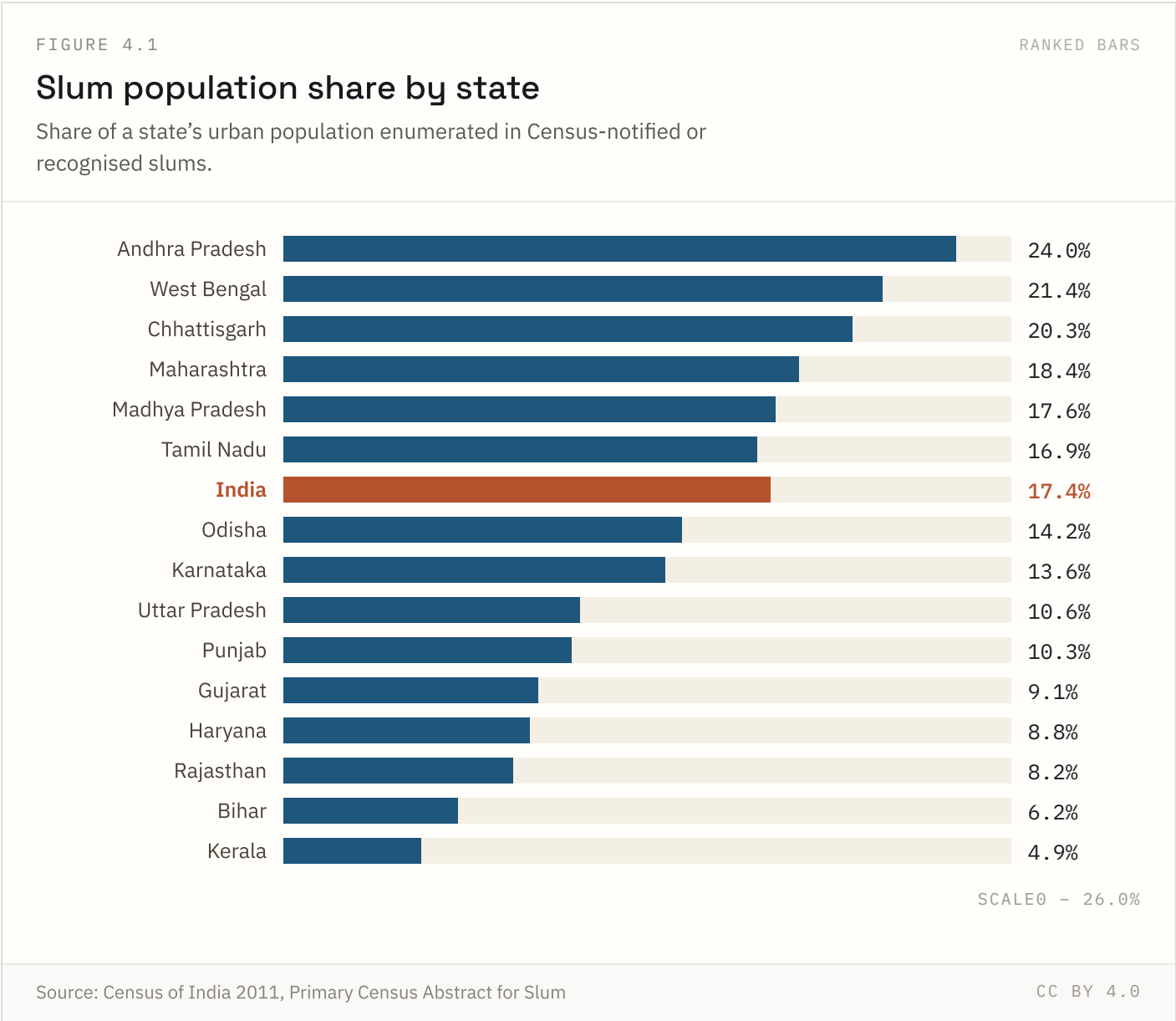
FIGURE 3.2 NATIONAL ONLY



Agriculture's share of the economy has fallen by more than half since 1951, from 53% to around 18%, even as output kept rising — the sector still employs a far larger share of the workforce than its economic share suggests. The 2021 uptick reflects the pandemic-era contraction of services and industry, not a farm revival.

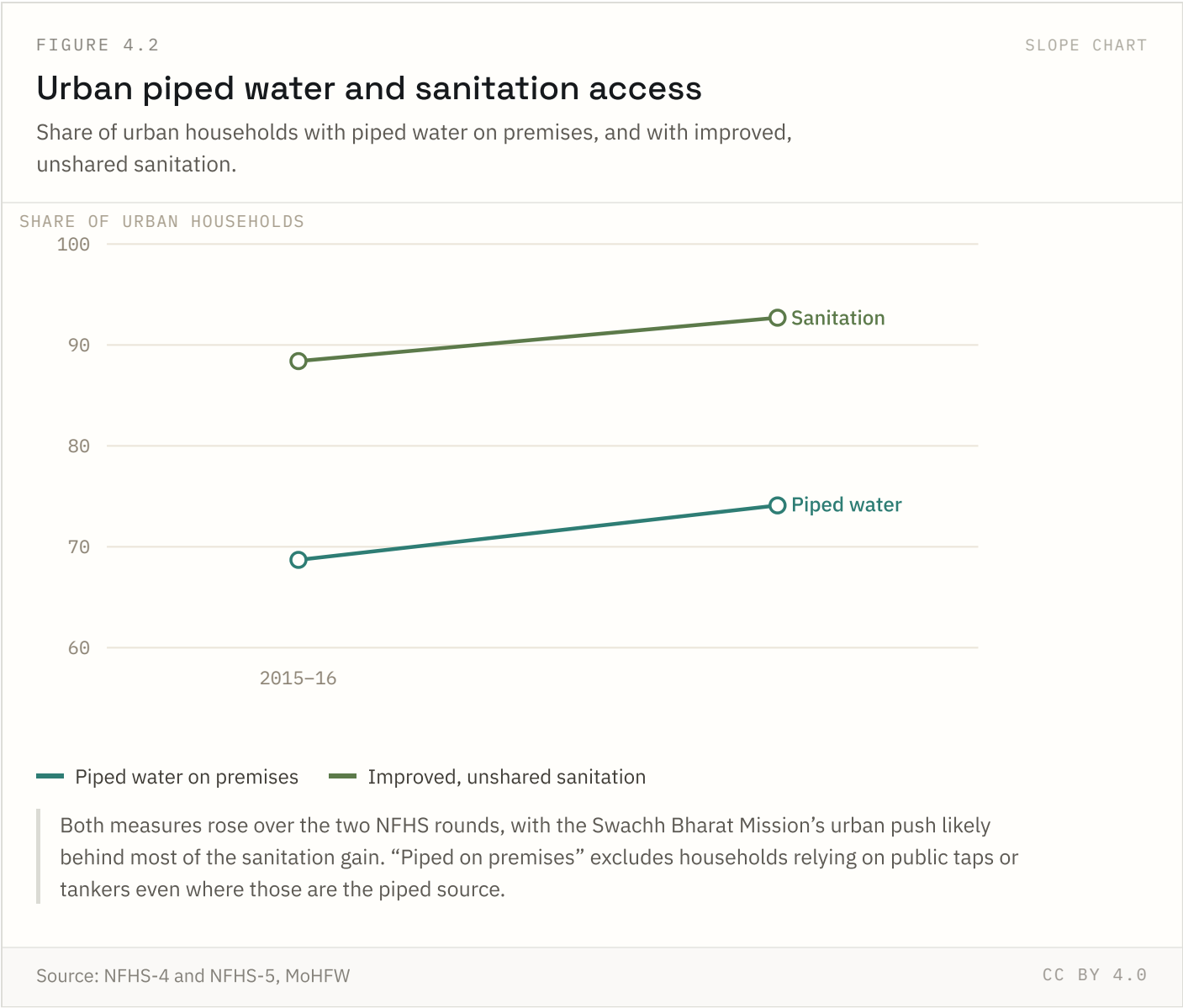
Urban development & housing

1 of 7 indicators in this chapter carry state-level detail; the rest are published at national level only.

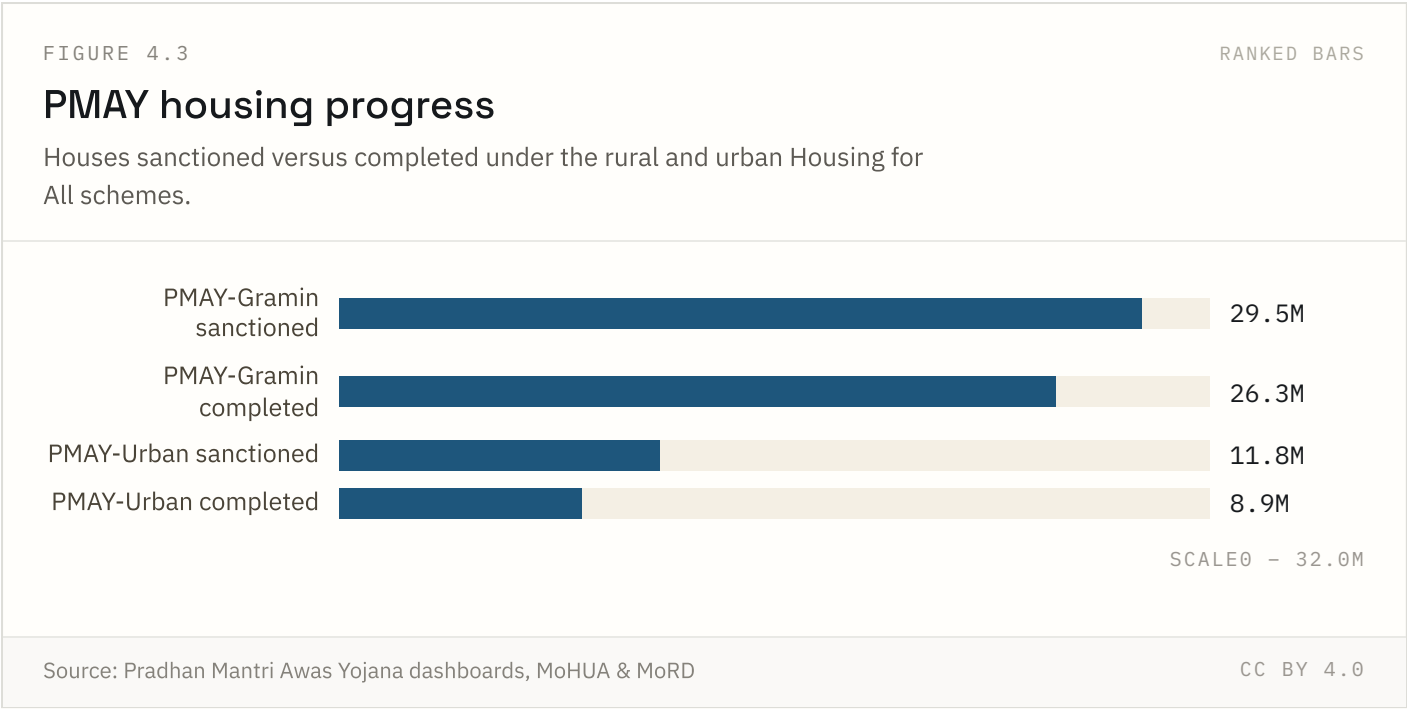


This is the last Census-based slum count available — no Census has been held since 2011, so growth in informal settlements over the past decade is not captured here. Definitions vary by state (notified, recognised and identified slums), which limits strict comparability.

Highest: **Andhra Pradesh** at 24.0%. Lowest: **Kerala** at 4.9%.



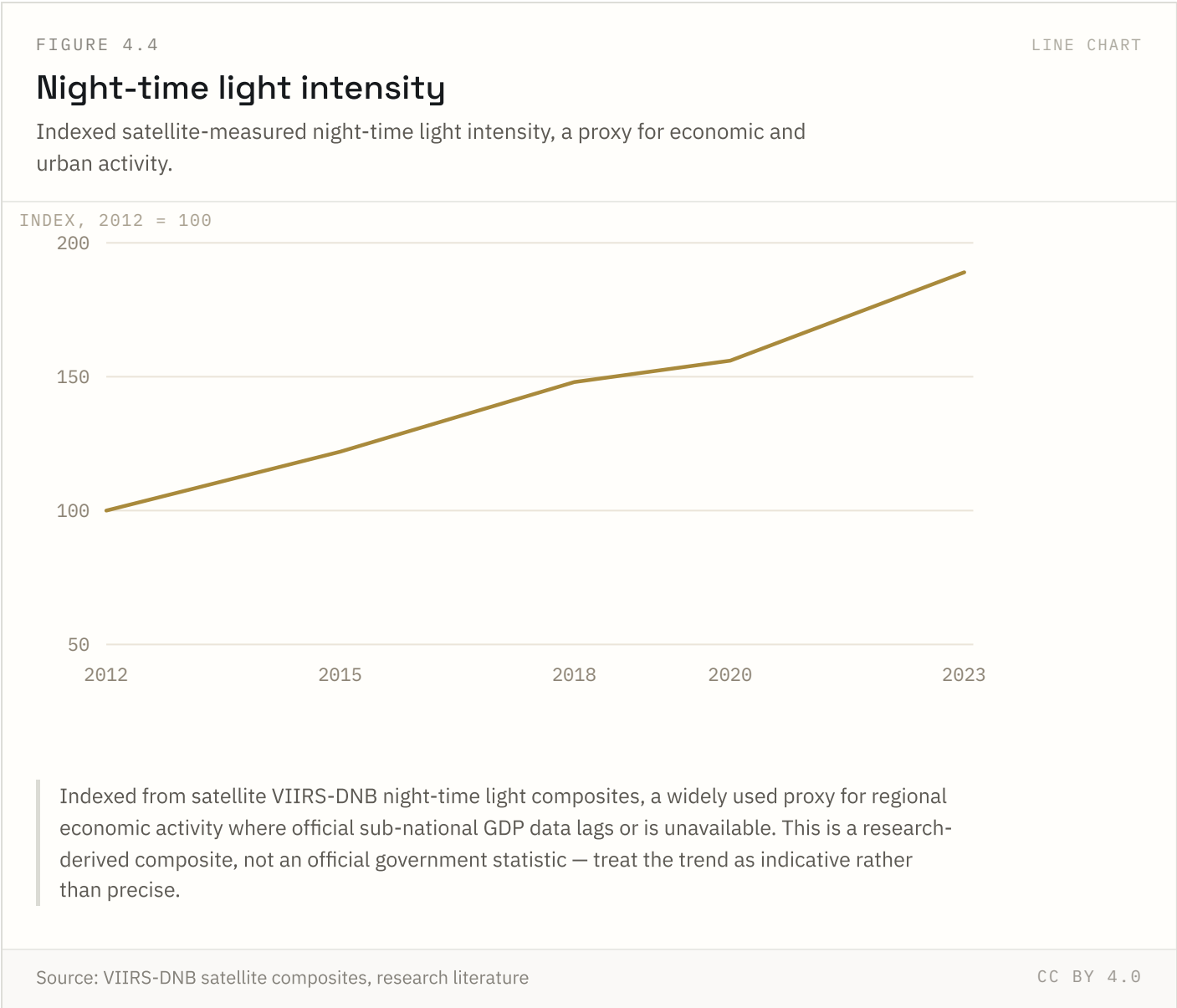
Both series improved between NFHS rounds. “Piped on premises” is stricter than “access to an improved water source” — it excludes households that rely on public taps, tankers or neighbours’ connections even if the underlying source is piped.



The rural scheme (PMAY-Gramin) is closer to its sanctioned target than the urban scheme (PMAY-Urban), where land acquisition and construction timelines run longer. “Completed” means handed over to the beneficiary, not merely under construction.

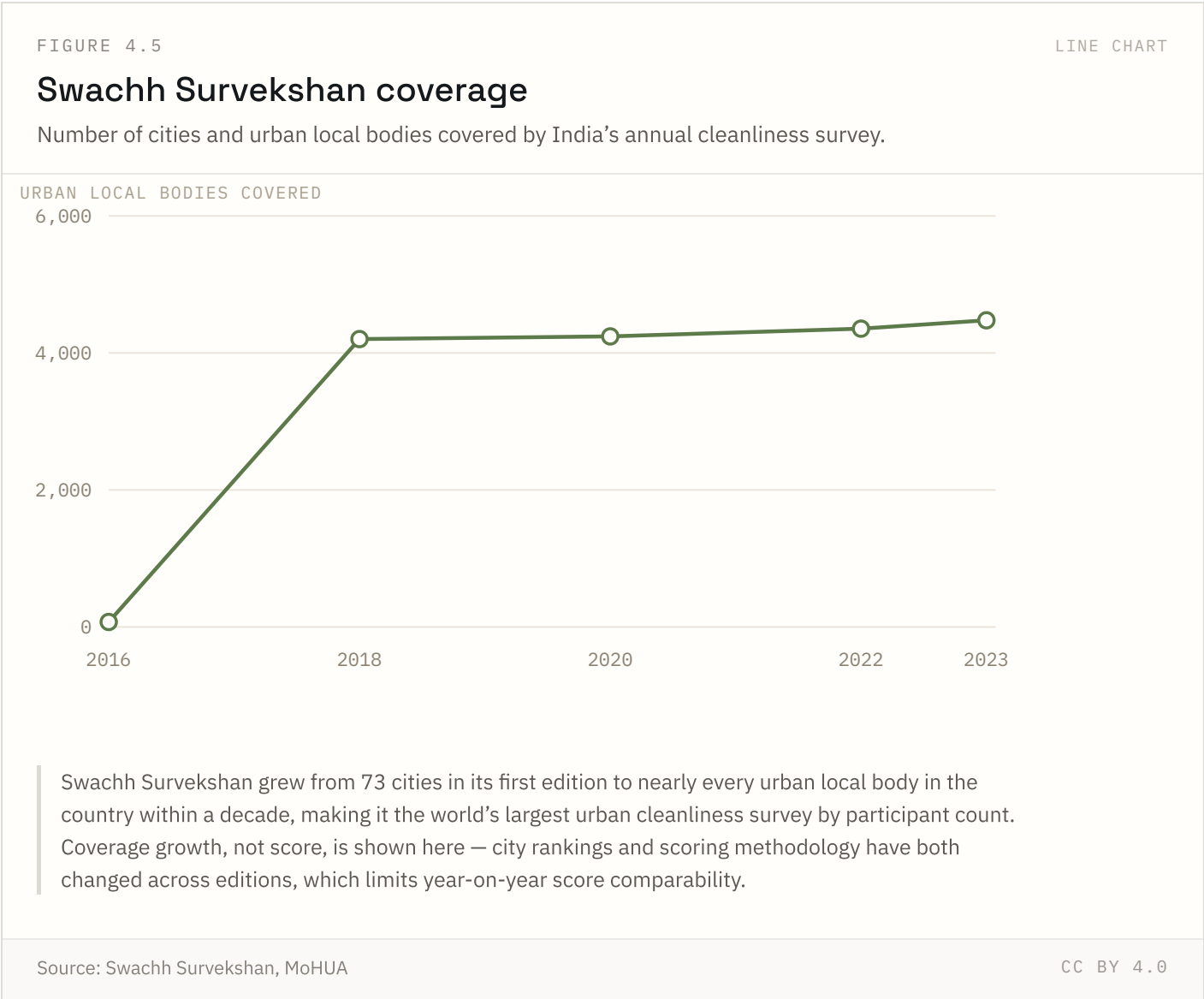
Source: Pradhan Mantri Awas Yojana dashboards, MoHUA & MoRD Coverage: Cumulative to 2025 Updated: 18 Aug 2026

FIGURE 4.4 NATIONAL ONLY



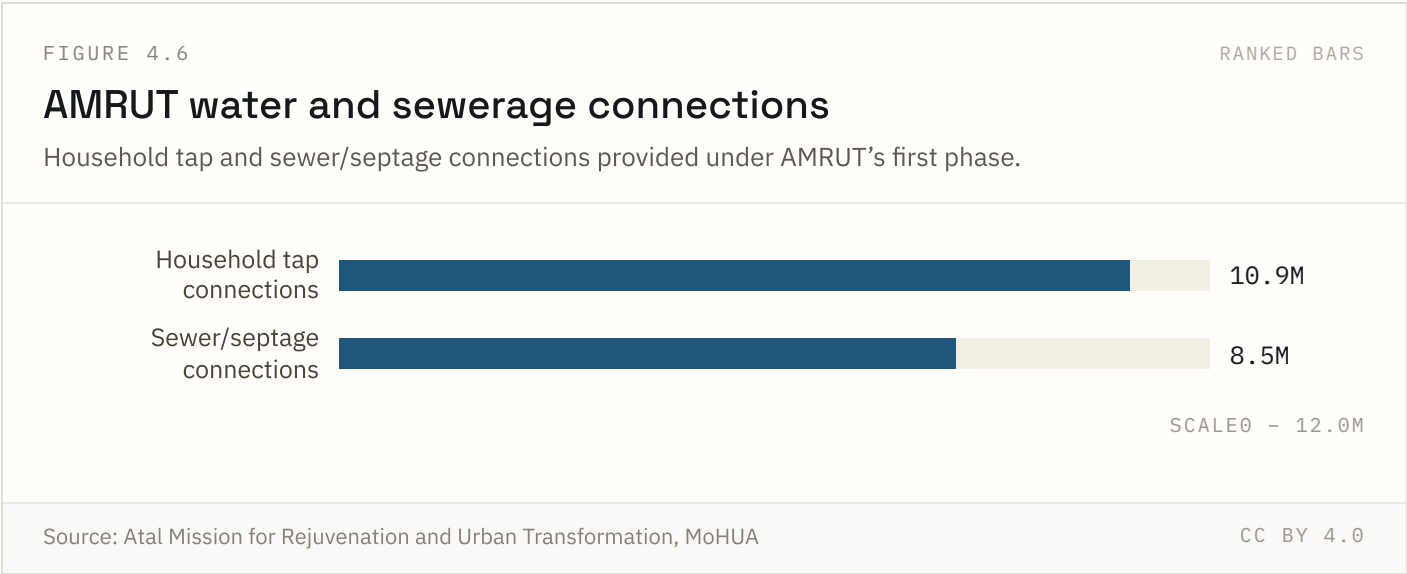
Night-time lights are used by researchers as a fast, sub-national proxy for economic growth where official data is thin or delayed. It captures electrified activity, not output directly, so it can overstate growth in areas that simply gained grid access rather than economic output.

FIGURE 4.5 NATIONAL ONLY



The survey has become close to universal across urban India, which is itself a policy achievement independent of any city’s score. Scoring methodology and weightings have changed across editions, so a city’s rank in different years is not a clean apples-to-apples comparison.

FIGURE 4.6 NATIONAL ONLY

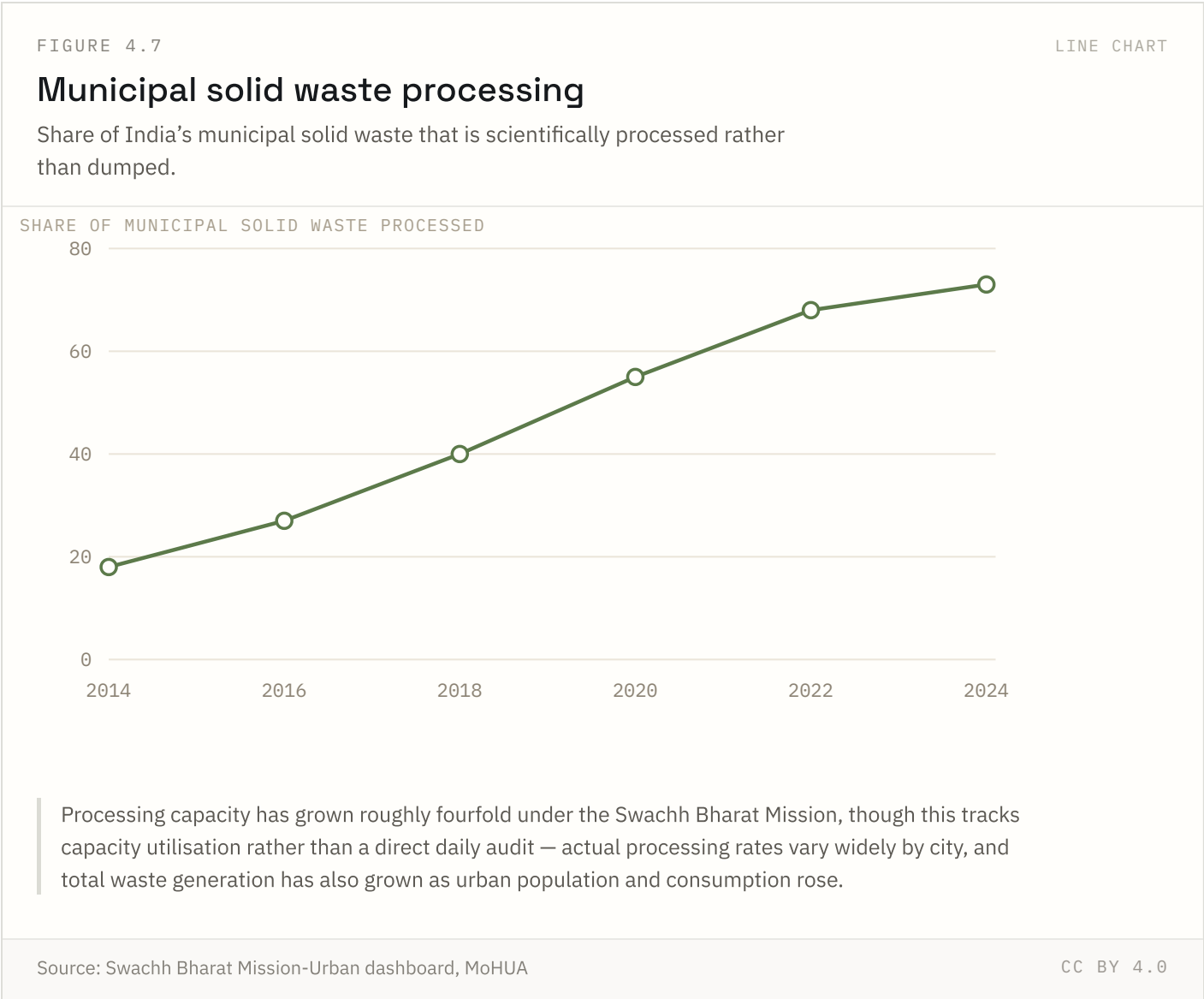


AMRUT 1.0 fell short of its full connection targets by its 2021 close; AMRUT 2.0 (launched 2021) aims at universal coverage. Sewerage lagged water supply throughout the scheme, reflecting the higher cost and longer build time of underground network projects.

Source: Atal Mission for Rejuvenation and Urban Transformation, MoHUA Coverage: 2015–2021 (AMRUT 1.0)

Updated: 18 Aug 2026

FIGURE 4.7 NATIONAL ONLY



Processing capacity has grown roughly fourfold since 2014, but total waste generation has also risen with urban population and consumption growth — a rising processed share does not necessarily mean a falling absolute quantity dumped in landfills.

Source: Swachh Bharat Mission-Urban dashboard, MoHUA Coverage: 2014–2024 Updated: 18 Aug 2026

Methodology and limitations

Every figure in this report is reproduced from a published source without adjustment. India Insight Lab does not estimate, model or interpolate the values shown. Each chart carries the source it came from, the years it covers and the date the series was last updated.

Three limitations apply to everything above, and are stated rather than worked around:

- **Geographic depth.** Indicators reach national and, for 5 of 24 series, state level. District and city figures are not included because comparable data is not yet integrated.
- **Comparability across sources.** Series drawn from different sources use different definitions, reference periods and sampling designs. They are shown side by side but are not adjusted onto a common basis.
- **Currency.** Update dates vary by series; the most recent revision date is shown beneath each figure. Some official series lag by several years.

Nothing in this report is projected. Where a series ends, the report ends with it. No forecast, scenario or extrapolation has been added to extend a trend beyond its published data.

References

22 sources used in this report.

1	Atal Mission for Rejuvenation and Urban Transformation, MoHUA
2	CPCB National Air Quality Monitoring Programme
3	Census of India 2011, Primary Census Abstract for Slum
4	Census of India; Saubhagya dashboard, Ministry of Power
5	Central Electricity Authority (CEA)
6	Central Ground Water Board (CGWB)
7	Global Carbon Budget
8	IQAir World Air Quality Report 2024
9	India Meteorological Department
10	India Meteorological Department, Statement on Climate of India
11	India State of Forest Report, Forest Survey of India
12	MNRE Physical Progress Report, Nov 2025
13	MNRE; CEA installed capacity reports
14	Ministry of Agriculture

15	NFHS-4 and NFHS-5, MoHFW
16	National Accounts Statistics, MoSPI
17	PM Surya Ghar Muft Bijli Yojana, Ministry of New & Renewable Energy
18	Pradhan Mantri Awas Yojana dashboards, MoHUA & MoRD
19	State drought declarations, compiled per the Manual for Drought Management
20	Swachh Bharat Mission-Urban dashboard, MoHUA
21	Swachh Survekshan, MoHUA
22	VIIRS-DNB satellite composites, research literature

Source URLs, licence terms and retrieval dates are not yet recorded for every series. Adding them is scheduled work, and until it is done these references name the publication rather than link to it.